

Entrepreneurship

Dr: Ahmed Mahran

Code: AHU20XX

Stage 2 – Full Project (Full Business Plan + MVP)
Dalla3 Kershak University Food Application

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1. Stage 1 Improvements

Identified Weak Points in Stage 1

- **Lack of Financial Depth:** Stage 1 lacked a structured 3-year projection and break-even analysis.
- **Vendor Onboarding Details:** The strategy to convince traditional, low-tech vendors to adopt a digital dashboard was under-developed.
- **Scalability Roadmap:** The plan focused heavily on ECU but lacked a structured vision for cross-campus expansion.

Improvements & Refinements

- **Problem Clarity:** The problem is not just "waiting in line"; it is a systemic bottleneck that caps vendor revenue during rush hours and causes a direct opportunity cost for students' academic performance.
 - **Refined Value Proposition:**
 - *For Students:* "Buy back your time. Skip the line, every time."
 - *For Vendors:* "Serve 30% more customers during the rush hour without hiring more staff."
 - **Differentiation:** Unlike delivery apps (Talabat), Dalla3 Kershak focuses purely on on-campus pickup, eliminating delivery logistics, fees, and campus entry restrictions.
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2. In-depth Market Analysis

SWOT Analysis

- **Strengths:** Hyper-localized solution, captive target audience, zero delivery costs/logistics, first-mover advantage in Egyptian universities.
- **Weaknesses:** Highly dependent on university infrastructure (internet availability), reliance on vendor compliance during high-stress hours.
- **Opportunities:** Expansion to other private universities in Greater Cairo, integration with university ID cards for payments, adding new services (e.g., printing center queues).
- **Threats:** Traditional vendors refusing technological adoption, university administration implementing their own restrictive food policies, or major delivery apps introducing "campus pickup" features.

PESTLE Analysis

- **Political:** University governance and administration regulations regarding commercial operations on campus.
 - **Economic:** Inflation in Egypt affects student purchasing power, making free-to-use apps and micro-commissions highly favorable over expensive delivery fees.
 - **Social:** Gen Z students demand frictionless, instant digital experiences. There is zero tolerance for manual, slow processes.
 - **Technological:** High smartphone penetration and growing comfort with digital wallets (Vodafone Cash, InstaPay) among Egyptian youth.
 - **Legal:** Compliance with the Central Bank of Egypt (CBE) regulations regarding digital wallets and e-payments.
 - **Environmental:** Significant reduction in paper receipts and reduced food waste through predictive preparation algorithms.
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3. Marketing & Sales Strategy (Go-to-Market)

The 7Ps of Marketing

- **Product:** A dual-sided digital ordering ecosystem (Student App + Vendor Dashboard).
- **Price:** Free for students. Vendors pay a 5 EGP micro-commission per order.
- **Place:** Implemented physically on the ECU campus and accessed digitally via smartphones.
- **Promotion:** Targeted on-campus activations, QR codes at kiosks, and student union partnerships.
- **People:** Driven by a passionate student tech team; utilized by tech-savvy students and trained cafeteria staff.
- **Process:** An optimized, 4-step workflow: Browse -> Schedule -> Pay -> Instant Pickup.
- **Physical Evidence:** Branded "Dalla3 Kershak Express Pickup" signs at participating vendors.

Digital Marketing & Launch Plan

- **Social Media Strategy:**

- **TikTok/Reels:** Short, relatable, humorous skits showing the pain of waiting in line versus the ease of using the app.
 - **Facebook Groups:** Announcements and promo codes in the official ECU student groups.
 - **Influencer Strategy:** Partnering with Student Union leaders and popular campus figures as brand ambassadors to drive initial trust and adoption.
 - **Campus Launch Activation:** Setting up physical booths during the first week of the semester.
 - ***Offer:*** "Deposit 50 EGP cash into your digital wallet today, get 55 EGP in app credit."
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4. Operations Plan

Product Development Roadmap (12 Weeks)

- **Weeks 1–4: Core MVP Development** (Authentication, UI implementation, Menu catalog, Cart logic).
- **Weeks 5–8: Advanced Features** (Digital Wallet logic, Vendor Dashboard, Socket.io real-time order tracking).
- **Weeks 9–10: QA, Load Testing** (simulating 12 PM rush), and bug fixing.
- **Weeks 11–12: Vendor Onboarding, staff training, and Soft Launch** with 2 anchor cafeterias.

Tech Stack

- **Frontend:** React.js (Mobile-first for students, Tablet-optimized for vendors).
- **Backend:** Node.js + Express.js (High concurrency for rush hours).
- **Database:** PostgreSQL (Financial transactions/Orders) + MongoDB (Flexible menu catalogs).
- **Real-Time:** WebSockets (Socket.io).
- **AI/Machine Learning:** Python (Scikit-learn) for predictive order preparation estimation.

Team Structure

- **CEO / Project Manager:** Strategic partnerships, vendor acquisition, and project timeline oversight.

- **CTO / Lead Backend Engineer:** Architecture, APIs, database management, and security.
- **Lead Frontend Engineer:** React.js development and state management.
- **UI/UX Designer:** Interface design, user research, and wireframing.
- **Data / AI Engineer:** Development of the dynamic preparation time algorithm.
- **Marketing & Operations Lead:** Social media, on-campus activations, and customer support.
- **QA Engineer:** End-to-end testing and performance load testing.

5. Financial Plan

Projected Income Statement (Summary - 3 Years)

Assumptions: Academic year = 200 active days. Average order commission = 3 EGP (lower, more realistic at early stage).

<i>Metric</i>	<i>Year 1 (1 Campus)</i>	<i>Year 2 (2 Campuses)</i>	<i>Year 3 (3 Campuses)</i>
<i>Daily Orders</i>	<i>50</i>	<i>120</i>	<i>250</i>
<i>Gross Commission Revenue</i>	<i>30,000 EGP</i>	<i>72,000 EGP</i>	<i>90,000 EGP</i>
<i>Ad / Promoted Listing Revenue</i>	<i>5,000 EGP</i>	<i>10,000 EGP</i>	<i>15,000 EGP</i>
<i>Total Revenue</i>	<i>35,000 EGP</i>	<i>82,000 EGP</i>	<i>100,000 EGP</i>
<i>Hosting & Tech Costs</i>	<i>(10,000 EGP)</i>	<i>(15,000 EGP)</i>	<i>(20,000 EGP)</i>
<i>Marketing & Activations</i>	<i>(8,000 EGP)</i>	<i>(12,000 EGP)</i>	<i>(18,000 EGP)</i>
<i>Salaries & Operations</i>	<i>(12,000 EGP)</i>	<i>(25,000 EGP)</i>	<i>(35,000 EGP)</i>
<i>Net Profit (Before Tax)</i>	<i>5,000 EGP</i>	<i>30,000 EGP</i>	<i>27,000 EGP</i>

6. Risk Analysis & Mitigation

Risk Area	Specific Risk	Mitigation Strategy
Operational	Vendors ignore the tablet during peak rush hour.	Design a massive, simplified UI (one-tap "Accept"). Deploy team members to shadow vendors during the first week.
Technical	Server crash during the 12:00 PM break surge.	Utilize AWS Auto-Scaling and conduct rigorous load-testing before launch.
Market	Students refuse to deposit cash into the digital wallet.	Offer a 10% credit bonus on initial deposits. Ensure instant refund policies.
Financial	Running out of cash before reaching critical mass.	Maintain a highly lean startup model; leverage free university channels (Facebook groups) for marketing.

7. Social & Environmental Impact

- **Impact on Students:** Drastically reduces daily stress. Reclaims an average of 2-3 hours per week per student, allowing for better study habits, socialization, and academic focus.
 - **Impact on Vendors:** Formalizes campus micro-economies. Teaches digital literacy to traditional cafeteria staff and increases their daily earning potential without requiring physical expansion.
 - **Sustainability:** The digital receipt and order management system eliminates thousands of paper slips daily. Furthermore, predictive ordering helps vendors manage ingredient prep, reducing daily food waste.
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8. MVP (Minimum Viable Product) Details

Functional Prototype Scope

The MVP will focus strictly on the core transaction loop.

- **In Scope:** Digital menu browsing, cart management, wallet payment, real-time status tracking, and the vendor command center.
- **Out of Scope for MVP:** AI predictive prep-time (will use static estimates initially), external gateway integrations (will rely on physical cash-to-digital wallet top-ups at booths).

User Flow (Step-by-Step)

1. **Discover:** Student opens web app, selects "ECU Main Cafeteria."
2. **Order:** Adds a Chicken Shawarma to cart.
3. **Schedule:** Selects pickup time: "11:45 AM".
4. **Pay:** Confirms order; 45 EGP is deducted from the digital wallet.
5. **Prep:** Vendor tablet dings. Vendor taps "Accept."
6. **Pickup:** At 11:45, student receives notification "Ready!" and walks directly to the dedicated "Dalla3 Kershak" pickup counter.